$$J_p \ddot{\theta} = \sum F_r$$

$$J_p \ddot{p} = l_p \cdot (F_L - F_f) + \cancel{l_p \cdot m_p \cdot g} - \cancel{l_p \cdot m_p \cdot g}$$

$$J_p \ddot{p} = l_p \cdot k_t \cdot (V_L - V_f)$$

$$J_p \ddot{p} = l_p \cdot k_t \cdot V_d$$

$$J_e \ddot{\varphi} = l_h (F_f + F_v) \cdot \cos(p) + l_c \cdot F_{gL} \cdot \cos(\varphi) - l_h (F_{gf} + F_{gL}) \cdot \cos(\varphi)$$

$$= l_h k_t \cdot V_s \cdot \cos(p) + l_c m_c g \cos(\varphi) - 2 l_h m_p g \cos(\varphi)$$

$$= \underbrace{(l_c m_c g - 2 l_h m_p g)}_{L_2} \cos(\varphi) + \underbrace{l_h k_t V_s}_{L_3} \cos(p)$$

$$J_\lambda \ddot{\lambda} = (F_f + F_v) \cdot l_h \cdot \sin(p) \cos(\varphi)$$

$$= \underbrace{l_h k_t V_s}_{L_4} \sin(p) \cos(\varphi)$$

Problem 3

$$\ddot{p} \approx \ddot{p}_0 + \frac{\partial \ddot{p}}{\partial p}(0)(p)$$

$$= \frac{l_p k_t}{J_p} \cdot V_a + 0$$

$$= \frac{l_p k_t}{2 m_p l_p^2} V_a = \left(\frac{k_t}{2 m_p l_p}\right) V_a$$

k_1

$$J_e \ddot{\theta} \approx \ddot{\theta}_0 + \frac{\partial \ddot{\theta}}{\partial \theta}(\theta_0)\theta + \frac{\partial \ddot{\theta}}{\partial p}(p_0)p$$

$$J_e \ddot{\theta} = l_2 + l_h k_t V_s + 0 + 0$$

$$= \left(l_c m_c g - 2 l_h m_p g\right) + l_h k_t \tilde{V}_s + l_h k_t V_{s,0}$$

$$V_{s,0} = -\frac{L_2}{L_h k_t} = \frac{(2 L_h m_p - L_c m_c) g}{L_h k_t}$$

$$J_e \ddot{\varepsilon} = \cancel{L_2} + L_h k_t \tilde{V}_s + \cancel{L_h k_t} \cdot \left(-\frac{\cancel{L_2}}{\cancel{L_h k_t}} \right)$$

$$\ddot{\varepsilon} = \frac{k_t}{m_c L_c^2 + 2 m_p L_h^2} \tilde{V}_s$$

$$V_{s,0} \cdot k_t = -\frac{L_2}{L_3}$$

$$k_t = -\frac{L_2}{L_3} \cdot V_{s,0}$$

$$\ddot{\lambda} \approx \ddot{\lambda}_0 + \frac{\partial \ddot{\lambda}}{\partial P}(P_0) P + \frac{\partial \ddot{\lambda}}{\partial \varepsilon}(\varepsilon_0) \varepsilon$$

$$k_t = -\frac{L_2 V}{L_h \cdot k}$$

$$k_t = \sqrt{\frac{L_2 V_{s,0}}{L_h}}$$

$$\approx 0 + \frac{L_h k_t V_{s,0} \cos(P_0) \cos(\varepsilon_0) P}{J_\lambda} + 0$$

$$\ddot{\lambda} = \frac{L_h k_t V_{s,0} P}{J_\lambda}$$

$$V_{s,0} = -\frac{L_2}{L_3}$$

$$k_t = -\frac{L_2}{L_h \cdot V_{s,0}}$$

Problem 3

$$\ddot{P} = k_1 V_d$$

$$\dddot{P} = k_1\left(k_{PP}(P_c - P) - k_{Pd}\dot{P}\right)$$

$$\mathcal{L}\{P\} = P$$

$$s^2 P = k_1 k_{PP}(P_c(s) - P) - k_1 k_{Pd} s P$$

$$P = \frac{k_1 k_{PP}}{s^2 + k_1 k_{Pd} s + k_1 k_{PP}}$$

$$G(s) = \frac{P(s)}{P_c(s)} = \frac{k_1 k_{PP}}{s^2 + k_1 k_{Pd} s + k_1 k_{PP}}$$

Problem 4

$$s^2 + K_1 K_{Pd} s + K_1 K_{PP}$$

$$s^2 + K_1 K_{Pd} s + K_1 K_{PP} = 0$$

$$\lambda = \frac{-K_1 K_{Pd} \pm \sqrt{K_1^2 K_{Pd}^2 - 4}}{2}$$

$$(s - \lambda_1)(s - \lambda_2) = 0$$

$$s^2 + s(-\lambda_1 - \lambda_2) + \lambda_1 \lambda_2 = 0$$

$$K_{PP} = \frac{\lambda_1 \lambda_2}{K_1} \qquad K_{Pd} = -\frac{\lambda_1 + \lambda_2}{K_1}$$

λ	
-1	$-1 \pm i$
-2	$-2 \pm 2i$
0	$\pm 3i$
1	
2	

$$x(t) = e^{(a+bi)t} x(0)$$

$$\ln(x(t)) = \ln((a+bi)t) + \ln(x(0))$$

$$= \ln(a+bi)t + \ln(x(0))$$